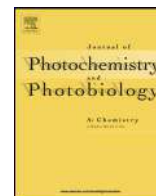




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Invited feature article

A new highly effective fluorescent probe for Al³⁺ ions and its application in practical samples



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ABSTRACT

A new 1,8-naphthalimide derivative, *N*-allyl-4-[3,3'-((2-aminoethyl)azanediyl) –bis(*N'*-(2-hydroxy-3-methoxybenzylidene)propanehydrazide)]-1,8-naphthalimide (**NBPN**), was designed, synthesized and evaluated as a chemosensor for Al³⁺ ion. The structure of probe **NBPN** was established by single crystal X-ray. Upon addition of different metal ions to the DMF/HEPES buffer (1:1, v/v, pH = 7.4) solution of **NBPN**, a significant selectivity toward Al³⁺ over other metal ions was observed. **NBPN** showed a good linearity with a correlation coefficient (*R*²) of 0.95 in the concentration range 3–11 μM. The respective detection limit of **NBPN** and the association constant for Al³⁺ were calculated to be 3.4×10^{-8} M and 1×10^4 M⁻¹. The stoichiometry and binding mode of probe **NBPN** with Al³⁺ were studied by spectroscopic methods of Job plot, UV–vis titration, fluorescence titration, MS and ¹H NMR titration and the experimental data indicated 1:1 stoichiometric complex between **NBPN** and Al³⁺. Ultimately, **NBPN** was employed for the detection of Al³⁺ in practical water samples.

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1. Introduction

Recently, the development and applications of fluorescent chemosensors for the detection of environmentally-significant metal ions have received increasing attention due to their advantages of high sensitivity, selectivity, real-time monitoring with response and low detection limit and much attention has been focused on the generation of sensing devices for aluminum (III) [1–9]. Aluminum, as the most abundant metal on earth, has been found in natural waters and most biological tissues. Moreover, aluminum is widely used and dispersed in industry and environment such as food additives, packing materials and clinical drugs etc [10–13]. Nevertheless, aluminum is not an essential element and exhibits toxicity for living systems. As a matter of fact, an excessive amount of aluminum constitutes much harm to human body and plant growth [14–17]. In this sense, the development of efficient fluorescent probes to detect Al³⁺ especially at very low ions concentrations has aroused the interests of many scientists.

Due to their unique properties of quite good chemical stability, high photostability, strong fluorescence and a marked Stokes shift, 1,8-Naphthalimide and its derivatives are being applied in a

number of areas such as polymers [18], DNA photo cleavage [19], paints and plastics [20] Fluorescence sensors and switchers [21], potential photosensitive biologically units [22] etc. Owing to their abilities to alter UV–vis absorption and fluorescence properties, researchers working on sensors are increasingly introducing various substituents to the 4 and 5 positions of the naphthalic ring, thereby making 1,8-naphthalimide derivatives widely designed and used as key units in the development of colorimetric and fluorescence sensors [23–25]. Owing to their capability to form complexes with transition and lanthanide metal ions, Schiff bases constitute a special group of organic compounds [26–30]. Meanwhile, there were some reports of Shift base based simple, stable and effective fluorescent sensors synthesized to detect metal ions and anion that displayed selective responses [31,32].

Herein, we would like to report a new Al³⁺-selective fluorescent probe (**NBPN**) based on 1,8-naphthalimide, which exhibits a large fluorescence enhancement upon binding to Al³⁺ with high selectivity and sensitivity.

2. Experimental

2.1. Materials and general method

All chemicals and solvents used for the synthesis were reagent grade and used without further purification. 4-bromo-1,8-naphthalic anhydride, ethanediamine, allylamine, methyl acrylate,

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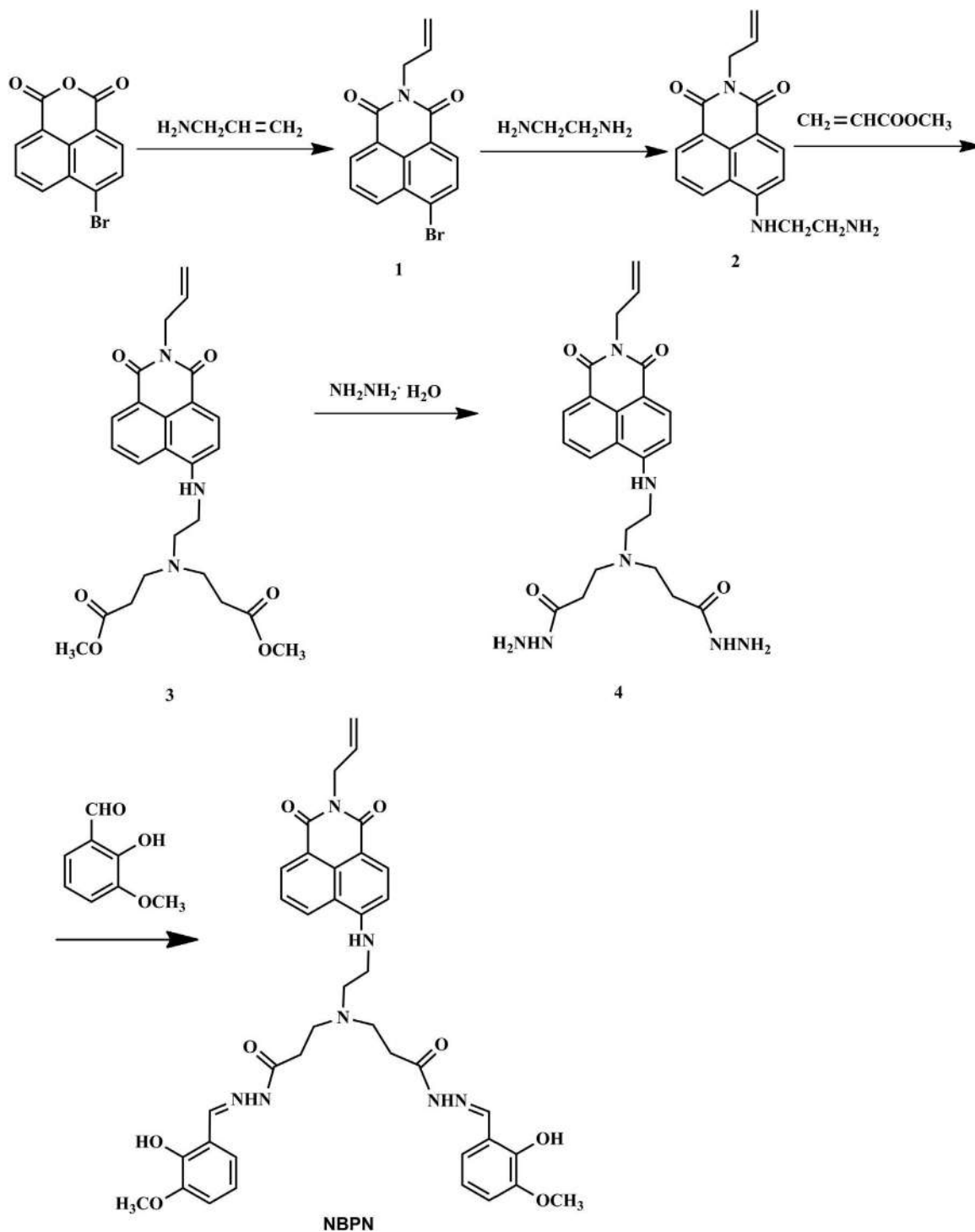
hydrazine hydrate, salicylaldehyde were bought from commercial suppliers. The stock solutions of the probe were obtained in DMF at a molar concentration of $1 \times 10^{-3} \text{ mol L}^{-1}$. HEPES buffer was prepared using double-distilled water.

C, H and N contents were determined using a Carlo Erba 1106 elemental analyzer. ^1H NMR spectra were recorded on a Varian VR300 MHz spectrometer with TMS as an internal standard. The IR spectra were recorded on a BRUKER FT-IR VERTEX 70 spectrometer in the range of $4000\text{--}400 \text{ cm}^{-1}$ using KBr pellets. Electronic spectra were taken on a Lab-Tech UV Bluestar spectrophotometer. The fluorescence spectra were recorded on

F97Pro fluorescence spectrofluorophotometer. The pH values were measured with a DELTA 320 pH meter. ESI-MS spectra were recorded on a Mass Spectrometer microTOF and MALDI-TOF mass spectra were collected on a BIFLEX III mass spectrometer. The melting points were measured by X-4 microscopic melting point apparatus. All the detections were carried out at 25°C .

2.2. Synthesis and characterization of fluorescence probe (NBPN)

The fluorescent probe (NBPN) was synthesized by a five-step reaction utilizing 4-bromo-1,8-naphthalimide anhydride as the



Scheme 1. The synthetic route of the NBPN.

starting material. The intermediate products **1–3** were prepared according to the reported method in previous literatures [33,34]. The synthetic route for the generation of **NBPN** is shown in Scheme 1.

2.2.1. Preparation of **4**

A suspension of product **3** (1 g, 2.14 mmol) in methanol (20 mL) was stirred under reflux at 60 °C. The solution was allowed to keep stirring until product **3** disappeared, after which it was cooled down to room temperature. Then, the solution was added to a mixture of hydrazine hydrate (8.13 g, 128.48 mmol) and methanol (10 mL). After a period of 24 h under room temperature, the resulting powder was washed with small amount of methanol several times in order to get rid of hydrazine hydrate, and finally dried in vacuo. The progress of the reaction was monitored by TLC (the TLC eluent was dichloromethane: methanol = 5:1, R_f = 0.4). Yield: 96%, mp: 182–183 °C. Anal. calcd. C 59.10; H 6.21; N 13.70%; found: C 59.31; H 6.10; N 13.92%.

2.2.2. Preparation of probe **NBPN**

A mixture of product **4** (0.5 g, 1.071 mmol), o-Vanillin (0.66 g, 4.34 mmol) and ethanol (20 mL) was stirred under reflux at 78 °C for 8 h. After the mixture was cooled down to room temperature, the precipitate produced was filtered, washed with absolute ethanol and dried to give 0.64 g yellow product. The progress of the reaction was monitored by TLC (the TLC eluent was dichloromethane: ethanol = 5:1, R_f = 0.5). Yield: 92%, mp: 187–188 °C. Anal. calcd. C 63.66; H 5.62; N 13.33%; found: C 63.60; H 5.51; N 13.52%. UV–vis (in DMF; λ /nm): 287, 327, 439. IR (KBr; ν /cm^{−1}): 1581.3s (C=O), 1648.9s (C=N), 1683.3s (C=C). ¹H NMR (DMSO-*d*₆, 400 MHz): δ (ppm) = 11.55 (d, 2H), δ (ppm) = 11.24 (s, 2H), δ (ppm) = 10.83 (d, 2H), δ (ppm) = 9.38 (s, 1H), δ (ppm) = 8.55 (m, J = 48 Hz, 1H), δ (ppm) = 8.31 (d, J = 7.2 Hz, 1H), δ (ppm) = 8.23 (d, J = 7.2 Hz, 1H), δ (ppm) = 8.20 (d, J = 9 Hz, 2H), δ (ppm) = 7.58 (d, J = 15.2 Hz, 1H), δ (ppm) = 7.48 (d, J = 13.6 Hz, 1H), δ (ppm) = 7.13 (m, J = 7.6 Hz, 1H), δ (ppm) = 6.89 (m, J = 7.6 Hz, 2H), δ (ppm) = 6.76 (m, J = 18.4 Hz, 1H), δ (ppm) = 5.91 (m, 1H), δ (ppm) = 5.09 (d, 2H), δ (ppm) = 4.59 (d, 2H), δ (ppm) = 3.78 (s, 6H), δ (ppm) = 3.47 (s, 2H), δ (ppm) = 2.85 (m, 8H), δ (ppm) = 2.45 (m, 2H). ¹³C NMR (DMSO-*d*₆, 400 MHz):

δ (ppm) = 173.60, 168.09, 163.79, 163.29, 162.95, 150.83, 148.22, 147.45, 147.01, 146.28, 141.12, 134.59, 133.77, 130.77, 129.78, 128.60, 128.47, 124.68, 121.31, 119.45, 119.05, 118.20, 116.47, 114.07, 113.09, 108.22, 104.35, 56.21, 51.65, 49.66, 49.12, 40.23, 39.95, 39.54, 32.82, 32.59, 30.55, 21.42, 11.87. HRMS (ESI, m/z): Calcd. 735.33, Found 736.3319 for $[M+H]^+$ (Figs. S1–S5).

2.3. X-ray crystallography of **NBPN**

The transparent single crystal of **NBPN** was obtained from the mixed solvent of methanol/THF by slow evaporation at room temperature. A suitable single crystal was mounted on a glass fiber, and the intensity data were collected on a Bruker APEX-II CCD diffractometer with graphite-monochromatized Mo K α radiation (λ = 0.71073 Å) at 296(2) K. The data reduction and cell refinement of crystal were performed with SMART and SAINT programs [35]. The structure and packing arrangements were obtained by direct computation methods and then refined by full-matrix least-squares technique on F^2 using SHELXTL software [36]. Basic crystal data, description of the diffraction experiment, and details concerning crystal data and refinement are summarized in Table 1, while the selected bond lengths and bond angles are given in Table 2. Fig. 1 gives the structure of **NBPN** along with its atomic numbering scheme; structural analysis reveals that **NBPN** was crystallized in a monoclinic space group P2(1)/c.

3. Results and discussion

The synthesis of probe **NBPN** based on 1,8-naphthalimide is outlined in Scheme 1. **NBPN** was confirmed using elemental analysis, NMR, MS, IR and X-ray crystal diffraction analytical techniques and its structure was found to be consistent with the proposed structure. **NBPN** is remarkably soluble in solvents such as DMF, DMSO, DCM and THF but insoluble in water, *n*-hexane.

3.1. Photophysical properties

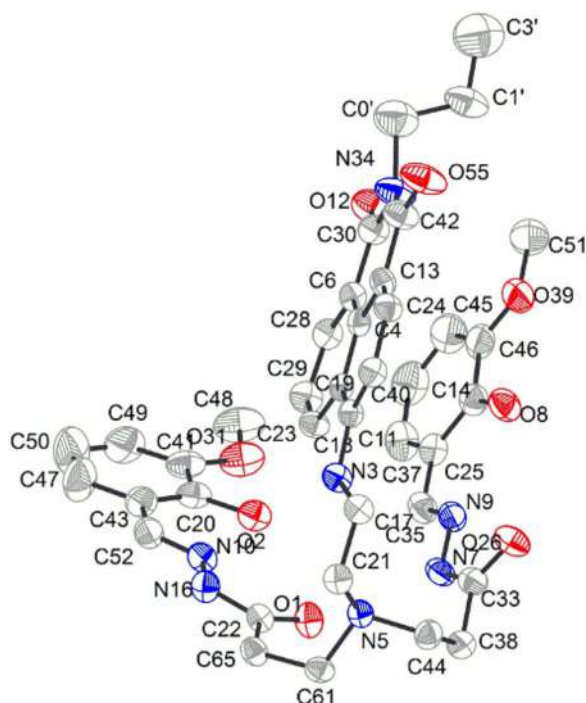
The fluorescence efficiency of the bis-1,8-naphthalimide was evaluated by measuring the fluorescent quantum yield Φ_F using

Table 1
Crystallographic data and structure refinement parameters for **NBPN**.

Complex	NBPN
Molecular formula	C ₃₉ H ₄₁ N ₇ O ₈
Molecular weight	735.79
Crystal system	Monoclinic
Space group	P2(1)/c
<i>a</i> (Å)	18.132(7)
<i>b</i> (Å)	15.592(6)
<i>c</i> (Å)	14.359(6)
α (°)	90
β (°)	112.644(7)
γ (°)	90(7)
<i>V</i> (Å ³)	3747(3)
<i>Z</i>	4
ρ_{calcd} (mg·m ^{−3})	1.304
Absorption coefficient (mm ^{−1})	0.093
<i>F</i> (000)	1552
Crystal size (mm)	0.40 × 0.38 × 0.30
θ range for data collection (°)	1.12 to 25.24
<i>h</i> / <i>k</i> / <i>l</i> (max, min)	−17 ≤ <i>h</i> ≤ 24, −18 ≤ <i>k</i> ≤ 18, −17 ≤ <i>l</i> ≤ 16
Reflections collected	15010
Independent reflections	5623
Refinement method	Full-matrix least-squares on F^2
Data/restraints/parameters	5623/32/502
Goodness-of-fit on F^2	0.847
Final <i>R</i> ₁ , <i>wR</i> ₂ indices [<i>I</i> > 2 σ (<i>I</i>)]	<i>R</i> ₁ = 0.0662, <i>wR</i> ₂ = 0.1590
<i>R</i> ₁ , <i>wR</i> ₂ indices (all data)	<i>R</i> ₁ = 0.1446, <i>wR</i> ₂ = 0.1932
Largest differences peak and hole (eÅ ^{−3})	0.241 and −0.220

Table 2
The relevant bond lengths (Å) and angles (°) for **NBPN**.

Bond lengths	O(1)–C(22)	1.226(4)	O(2)–C(20)	1.347(5)
	N(3)–C(11)	1.370(4)	N(3)–C(17)	1.453(4)
	N(5)–C(61)	1.463(4)		
	N(55)–C(21)	1.471(5)	N(5)–C(44)	1.470(4)
	N(7)–N(9)	1.370(4)	N(7)–C(33)	1.355(5)
	N(9)–C(35)	1.289(5)	O(8)–C(14)	1.342(5)
	N(10)–N(16)	1.370(4)	N(10)–C(52)	1.263(4)
	N(16)–C(22)	1.346(5)	O(12)–C(30)	1.229(4)
	C(30)–N(34)	1.388(5)	O(26)–C(33)	1.229(4)
	O(31)–C(48)	1.420(5)	O(31)–C(41)	1.375(6)
	N(34)–C(1)	1.519(13)	N(34)–C(42)	1.404(5)
	O(39)–C(46)	1.371(6)	N(34)–C(0')	1.52(2)
	C(42)–O(55)	1.221(5)	O(39)–C(51)	1.377(6)
Bond angles	C(11)–N(3)–C(17)	121.6(3)	C(61)–N(5)–C(44)	109.7(3)
	C(61)–N(5)–C(21)	112.4(3)	C(44)–N(5)–C(21)	110.7(3)
	C(33)–N(9)–N(7)	118.5(3)	C(35)–N(9)–N(7)	119.2(3)
	C(52)–N(10)–N(16)	118.3(3)	C(22)–N(16)–N(10)	119.8(3)
	O(12)–C(30)–N(34)	120.0(4)	C(30)–N(34)–C(0')	116.6(8)
	C(42)–N(34)–C(0')	116.8(9)	O(55)–C(42)–N(34)	119.4(4)
	N(9)–C(35)–C(25)	120.2(4)	N(10)–C(52)–C(43)	121.6(4)

**Fig. 1.** X-ray single crystal structure of **NBPN**.

Eq. (1). Herein, Φ_F was estimated from the absorption and fluorescence spectra of **NBPN**, the subscript s and r stand for the sample and reference, respectively. Φ_{ref} is the emission quantum

yield of the standard, A refers to the absorbance at the excitation wavelength. S represents the integrated emission band areas and n_{ref} and n_{sample} are the solvent refractive index of the standard and sample, respectively [37].

$$\Phi_F = \Phi_{ref} (S_{sample}/S_{ref}) (A_{ref}/A_{sample}) (n_{sample}/n_{ref})^2 \quad (1)$$

The standard used for the measurement of fluorescence quantum yield was *N*-butyl-4-butylamino-naphthalene imide ($\Phi = 0.81$ in ethanol) [38]. The photophysical properties of substituted 1,8-naphthalimides are known to depend mainly on the polarization of their chromophoric system [39]. Thereby, photophysical characteristics of **NBPN** in organic solvents with different polarity have been investigated. Its photophysical characteristics: absorption (λ_A) and fluorescence (λ_F) maxima, the extinction coefficient (ϵ), the Stokes shift ($\nu_A - \nu_F$), and quantum fluorescence yield (Φ_F) were shown in Table 3.

The ability of **NBPN** to emit absorbed light energy is characterized quantitatively by the quantum yield of fluorescence, Φ_F . Table 3 presents the quantum fluorescence yield Φ_F of **NBPN** is highly solvent dependent, which is obviously less in non-polar solvent than in polar solvent. This may be due to the existence of electron-releasing substituent at the C4 position of the 1,8-naphthalimide unit. Moreover, the Stokes shift is an important parameter, which indicates the difference in the properties and structure of the fluorophore between the ground state S_0 , and the first excited state S_1 . The Stokes shift of **NBPN** is in the region of 3198 (DMSO)–4748 (methanol) cm^{-1} . It is seen that the polarity of the organic solvents has great influence on the Stokes shift values and suggests larger differences in character and structures of the 1,8-naphthalimide in **NBPN** between the S_0 and S_1 states in methanol than in other solvents. The extinction coefficient (ϵ) of **NBPN** ranged from 5705 to 14373, which deduces that ϵ is larger in polar solvents than in the non-polar solvents. This great difference can be explained on the basis of the polarity of the organic solvents, which has an assignable influence on the ϵ of **NBPN**, which is turn relatively much more solvent dependent.

3.2. The fluorescence spectra titration of **NBPN** for Al^{3+}

The effect of pH on the fluorescence emission was investigated in DMF solution at different pH conditions ranging from 0 to 14.0 with **NBPN** concentration maintained at 10^{-3} M^{-1} . The pH values were maintained by HEPES buffer and NaOH and HCl solution was used to adjust the acidity of solutions [33]. As shown in Fig. 2, the fluorescence intensity was observed to be constant and relatively unaffected between pH 4.51–10.56, but decreased gradually in the strong acidic and basic conditions of 0.53–3.58 and 10.56–13.52. This indicates that **NBPN** is suitable as Al^{3+} fluorescent probe under neutral aqueous medium. According to the results, the following experiments were carried out in solution of pH = 7.4. However, the emission of Al^{3+} complex maintained fairly intense maximum emission between pH 6.55–8.53, while at pH > 8.53, the fluorescence intensity decrease may be due to the formation of Al

Table 3
Photophysical characteristics of **NBPN**.

Solvent	$\lambda_A(\text{nm})$	$\epsilon(1 \text{ mol}^{-1} \text{ cm}^{-1})$	$\lambda_F(\text{nm})$	$\nu_A - \nu_F (\text{cm}^{-1})$	Φ_F
DMSO	451	14,373	527	3198	0.25
Acetonitrile	436	11,448	520	3705	0.51
DMF	441	10,960	521	3482	0.27
Methanol	426	14,330	534	4748	0.04
Ethanol	451	9,028	528	3234	0.24
Acetone	436	12,388	514	3481	0.59
THF	434	10,683	505	3239	0.8
Dichloromethane	435	5705	510	3381	0.82

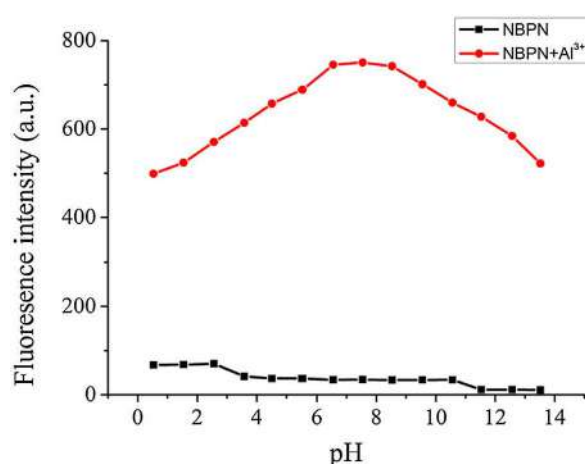


Fig. 2. Fluorescence spectra of probe **NBPB** in the absence and presence of Al^{3+} at various pH values, from top to bottom, pH: 0–14, plot of fluorescence intensity depending on the pH values (excitation at 350 nm, 25 °C).

(OH)₃ thereby reducing the concentration of Al^{3+} complex. This result reveals that probe **NBPB** exhibits satisfactory Al^{3+} sensing ability in the medium pH range.

A series of metal ions containing Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Al^{3+} , Pb^{2+} , Fe^{3+} , Ni^{2+} , Zn^{2+} , Cu^{2+} , Hg^{2+} , Ag^+ , Co^{2+} , Cr^{3+} , Mn^{2+} , Cd^{2+} , Fe^{2+} , Ga^{3+} and In^{3+} has been investigated to validate the selectivity of **NBPB** towards Al^{3+} [40]. The nitrate salts were used to prepare solutions of the metal ions. As given in Fig. 3a, when excited at 350 nm, **NBPB** showed weak fluorescence intensity at an emission wavelength at 527 nm but did not show obvious fluorescence changes upon the addition of various cations except Al^{3+} into stock solution of **NBPB**. However, a remarkable fluorescence enhancement of approximately 8-fold increase of the free probe **NBPB** was observed in the presence of Al^{3+} under the same condition. Only Al^{3+} generated evident fluorescence enhancement relative to the relevant ions. It seems that the reason for fluorescence enhancement is that **NBPB** has stronger binding affinity toward Al^{3+} than other coexistent metal cations investigated.

To utilize probe **NBPB** as an ion-selective fluorescence probe towards Al^{3+} , the effect of related metal ions was studied. The probe **NBPB** was measured with 20 μM of Al^{3+} by the addition of other relevant metal ions including Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Pb^{2+} , Fe^{3+} , Ni^{2+} , Zn^{2+} , Cu^{2+} , Hg^{2+} , Ag^+ , Co^{2+} , Cr^{3+} , Mn^{2+} , Cd^{2+} , Fe^{2+} , Ga^{3+} and In^{3+} of the same concentration [41]. As shown in Fig. 3b, the presence of

other cations did not appear to have any significant fluorescence signal interference effect on the **NBPB**- Al^{3+} , and the unique spectral and color changes were still retained as before. The results indicate that probe **NBPB** could be used as a selective fluorescent probe for detecting Al^{3+} in the presence of other competing metal ions and the detection of Al^{3+} by **NBPB** was insignificantly affected by these coexistent metal cations.

To investigate the influence of the addition of incremental concentrations of Al^{3+} on fluorescence intensity of probe **NBPB**, the fluorescence titration of **NBPB** toward Al^{3+} was carried out [13,42]. As revealed in Fig. 4, when an increasing concentration of Al^{3+} was added to the solution of 10 μM probe **NBPB**, the emission peak of **NBPB** at 528 nm was enhanced obviously. The fluorescence enhancement was due to the binding of Al^{3+} and probe, which reduced the electron-donating of nitrogen atom of Schiff base and inhibited the PET process [43]. The design of sensor **NBPB** was based on a classical photoinduced electron transfer (PET) principle. Our proposed sensor scaffold is made up of a *N*-allyl-4-amino-1,8-naphthalimide ring that acts as the fluorophore, a group of a nitrogen atoms anchoring the Schiff base groups that acts as the receptor, and an ethylene group that acts as the spacer. With such a design in place, there was facilitation of the retainment of the fluorescence properties of the sensor upon coordination with each of the metal ion, but a significant enhancement of the fluorescence in the case of Al^{3+} ion. Additionally, after 1.0 equiv. of Al^{3+} ions was added, the emission intensity of the systems was stabilized and underwent no further changes. These results indicate the formation of a 1:1 complex between **NBPB** and Al^{3+} .

According to the linear fitting of the fluorescence titration data, binding constant K was calculated based on the Benesi–Hildebrand Eq. (1) [8].

$$1/(F_x - F_0) = 1/(F_{\max} - F_0) + (1/K[C])(F_{\max} - F_0) \quad (1)$$

Herein, F_0 , F_x , and $F_{\max}$ are the emission intensities (at 350 nm) of the organic moiety considered in the absence of metal ion, at an intermediate metal ion concentration, and at a concentration of complete interaction, respectively, and K is the binding constant concentration [44].

The binding constant K_a of **NBPB**- Al^{3+} was evaluated to be $1 \times 10^4 \text{ M}^{-1}$ (error limits $\leq 10\%$) (Fig. 5). Moreover, the increased fluorescence intensity with the increasing amount of the Al^{3+} added (3–11 μM) demonstrates a good linearity with a correlation coefficient (R^2) of 0.95 [45]. The detection limit of **NBPB** for Al^{3+} was calculated to be $3.4 \times 10^{-8} \text{ M}$, using the equation: $C_{\text{DL}} = 3\delta/K$; where δ is the standard deviation of the blank solution and K is the

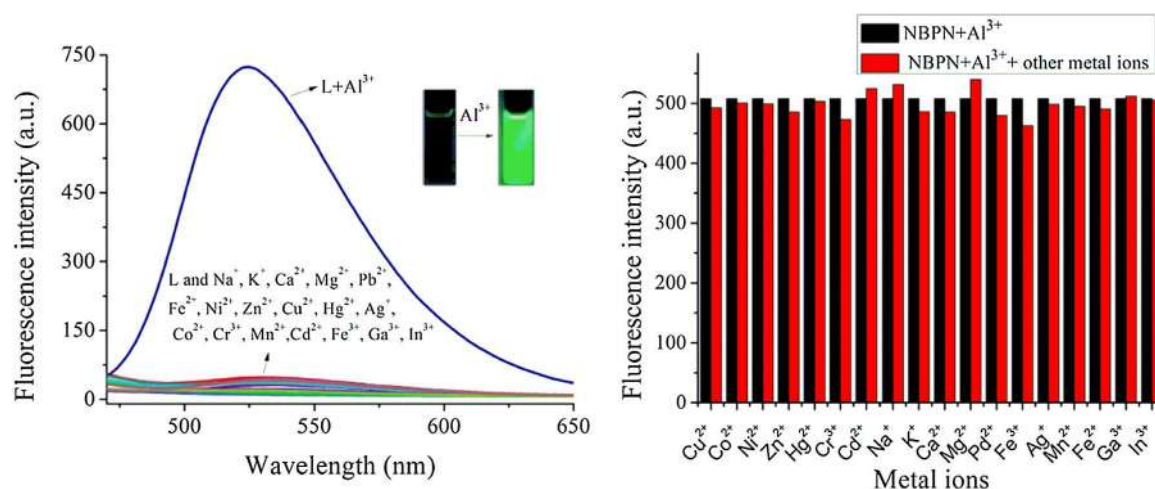


Fig. 3. a: Fluorescence spectra of **NBPB** (5 μM) upon the addition of various metal salts of Na^+ , K^+ , Ca^{2+} , Mg^{2+} , Pb^{2+} , Fe^{3+} , Ni^{2+} , Zn^{2+} , Cu^{2+} , Hg^{2+} , Ag^+ , Co^{2+} , Cr^{3+} , Mn^{2+} , Cd^{2+} , Al^{3+} , Fe^{2+} , Ga^{3+} and In^{3+} . Inset: color of **NBPB** and **NBPB** + Al^{3+} system under UV lamp (365 nm); b: The selectivity of **NBPB** for Al^{3+} in the presence of other metal ions.

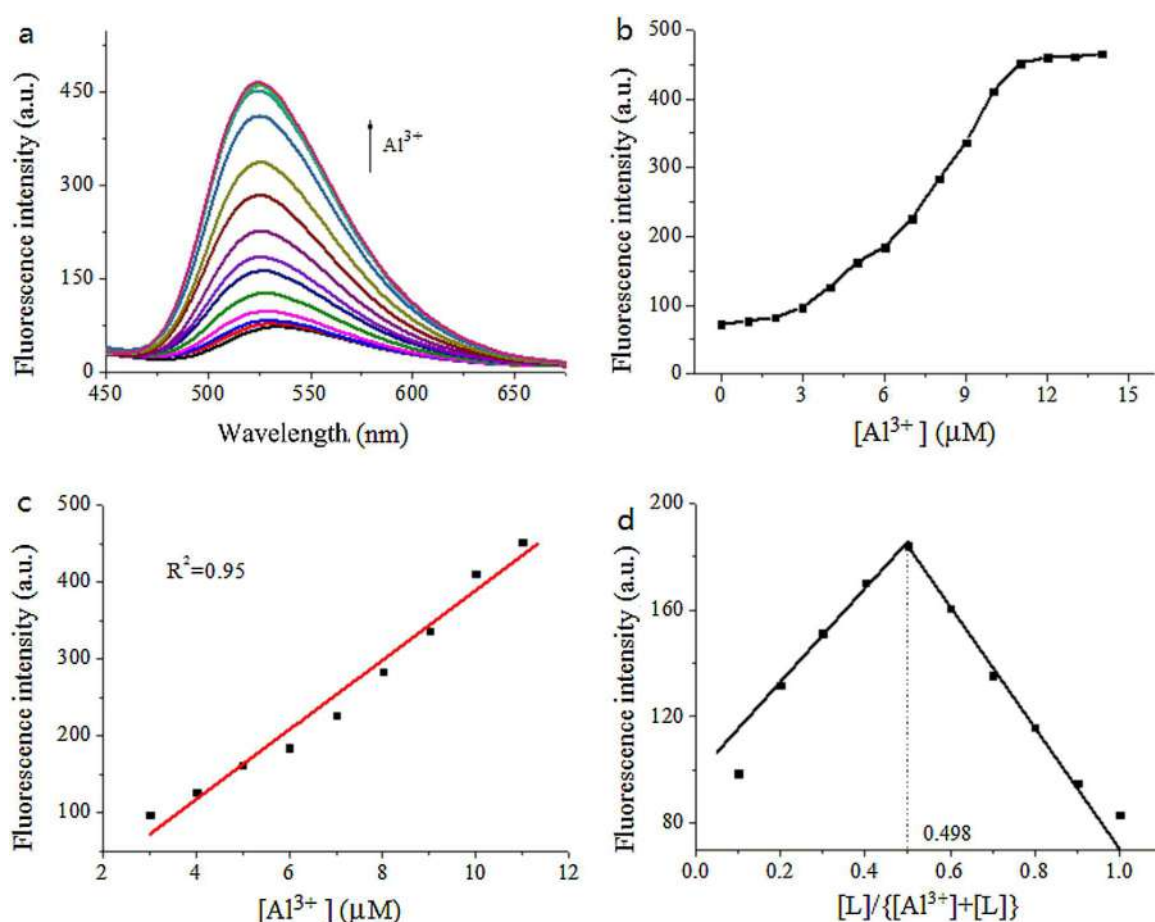


Fig. 4. a: Fluorescence spectra of NBPn (10 μM) upon addition of Al^{3+} in HEPES (pH = 7.4)/DMF (v/v, 1:1) at room temperature with an excitation of 350 nm; b: Fluorescence intensity at 528 nm of the sensor spots of NBPn upon addition of different amounts of Al^{3+} (0–14 μM); c: the linear proportionality of fluorescence intensity with increasing concentrations of Al^{3+} (3–11 μM); d: Job's plot for determining the stoichiometry of NBPn and Al^{3+} in HEPES (pH = 7.4) buffer/Ethanol (1:1, v/v) solution. Excitation wavelength was 350 nm.

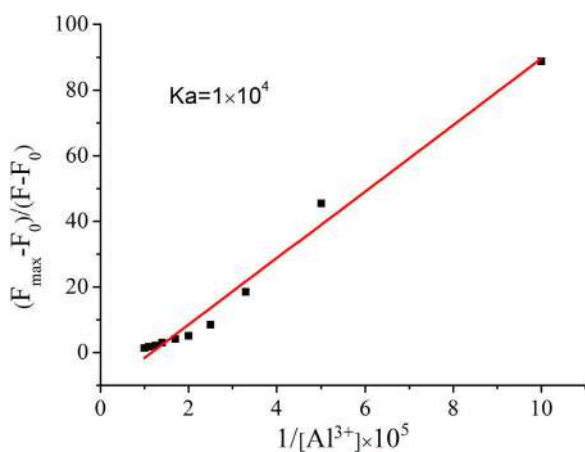


Fig. 5. Binding constant calculated of NBPn- Al^{3+} .

slope of the calibration curve. The plot of fluorescence intensity as a function of concentration of Al^{3+} ions suggests that NBPn can detect Al^{3+} even at submicromolar levels.

To further confirm the 1:1 stoichiometry between NBPn and Al^{3+} , Job's plot analysis was performed in aqueous media (HEPES, pH = 7.4) [46]. As recorded in Fig. 4, based on the changes in fluorescence intensity at 528 nm, Job's plot confirmed the formation of a 1:1 complex of NBPn with Al^{3+} and indicated that

the formation of NBPn- Al^{3+} resulted in significant fluorescence enhancement.

3.3. The UV-vis absorption of NBPn to Al^{3+}

With the gradual addition of Al^{3+} , the changes in the UV-vis spectra of NBPn was investigated in a HEPES (pH = 7.4) buffer/DMF (1:1, v/v) solution to research the binding mode of NBPn with Al^{3+} [47]. The changes in the absorption spectra of NBPn as a function of the concentration of Al^{3+} were depicted in Fig. 6.

NBPn exhibited a weak absorption band in the ultraviolet region. From the curve in Fig. 6a, it can be seen that the bands at 300 nm and 480 nm gradually increased upon the increasing concentration of Al^{3+} ion and these can be ascribed to the π - π^* transition of the benzene ring [48]. Simultaneously, the absorption peak at 375 nm (assigned to C=N group) gradually increased linearly upon the addition of increasing amount of Al^{3+} , suggesting the binding of NBPn- Al^{3+} complex. These significant changes display clear evidences of C=N involved in coordination to Al^{3+} ion and displayed that NBPn shows a very high sensitivity towards Al^{3+} . As given in Fig. 6b, as the Al^{3+} ion was gradually titrated, the absorbance showed negligible changes and achieved saturation at the ratio of about 1:1 between NBPn and Al^{3+} .

The UV-vis spectrum changes indicate that the formation of 1:1 stoichiometry complex between NBPn and Al^{3+} is in agreement with the conclusion of fluorescence spectra titration for Al^{3+} and Job plot measurements.

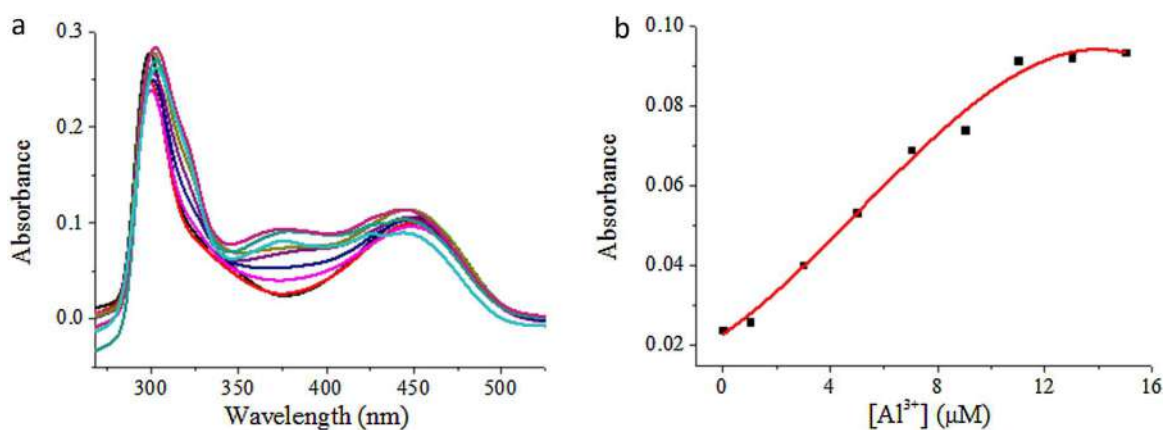


Fig. 6. a: Changes in the UV-vis spectra of **NBPB** (10 μM) in the media of HEPES (PH = 7.4)/DMF (v/v, 1:1) upon addition of various concentrations of Al^{3+} (0–15 μM); b: the absorbance changes at 375 nm upon the addition of Al^{3+} .

3.4. ^1H NMR titration

Besides the above experiments, ^1H NMR spectroscopy has been widely applied for confirming the binding mode, coordinative sites, and binding mechanism between compounds **NBPB** and metal ions [49,50]. So, the ^1H NMR titration spectra of **NBPB** in the absence and presence of Al^{3+} were conducted as shown in Fig. 7.

As shown in Fig. 7, in the ^1H NMR of the neutral **NBPB**, the proton signal of $-\text{OH}$ group at 11.56 ppm, after the addition of Al^{3+} neither disappeared nor shifted, suggesting that $-\text{OH}$ group might not be deprotonated and involved in coordination with Al^{3+} . In addition, the proton signals of the ethanediamine group at 3.47 and 2.51 ppm shifted obviously to upfield at 3.81 ppm and 2.84 ppm respectively, compared to the ^1H NMR spectrum of **NBPB**, which were attributed to the coordination between nitrogen atoms of ethanediamine group in **NBPB** and Al^{3+} . Moreover, with the addition of Al^{3+} , the proton peak of $\text{HC}=\text{N}$ as well as that of benzene rings were shifted upfield to the region of 0.6 ppm owing to the coordination between the N of Schiff base with Al^{3+} . By the information from ^1H NMR titration data, we could primarily confirm that the coordinative sites of **NBPB** for Al^{3+} was from the

nitrogen atoms of ethanediamine and Schiff base group, while the $-\text{OH}$ group did not participate in coordination with Al^{3+} , which agreed well with the conclusion of fluorescence titration and absorbance titration.

In order to further demonstrate the stoichiometry between **NBPB** and Al^{3+} ion, MALDI-TOF mass spectrometry was conducted. Mass peaks at m/z 762.3055 (calcd 762.32) corresponding to $[\text{NBPB} + \text{Al}^{3+}]^+$, and 736.3319 (calcd 736.33) corresponded to $[\text{NBPB} + \text{H}]^+$ were clearly observed when Al^{3+} was added to **NBPB**, which provided evidence for the formation of a 1:1 complex (Fig. S6).

All the results of Job's plot, ^1H NMR titration, fluorescence titration, absorbance titration and MS analysis accounted for that **NBPB** may chelate Al^{3+} through interactions with ethanediamine nitrogen and nitrogen of Schiff base group. There was also a new complex formation of **NBPB**- Al^{3+} (**NBPB**: Al^{3+} = 1:1), which involved a new interaction way that is different from the reported way of replacement of protons [51,52]. Additionally, the pH effect on **NBPB** + Al^{3+} sensor complex and the effect of counter anions of the salts of various metal ions have been researched. The data of experiment showed **NBPB** + Al^{3+} sensor complex with no such

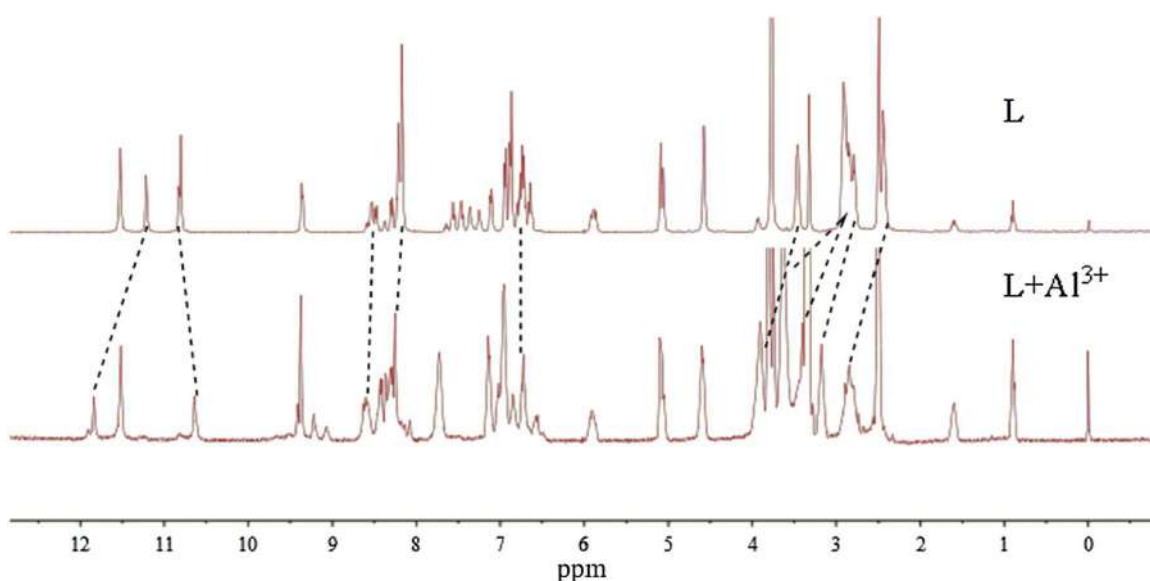


Fig. 7. ^1H NMR spectra of **NBPB** and with Al^{3+} in $\text{DMSO}-d_6$.

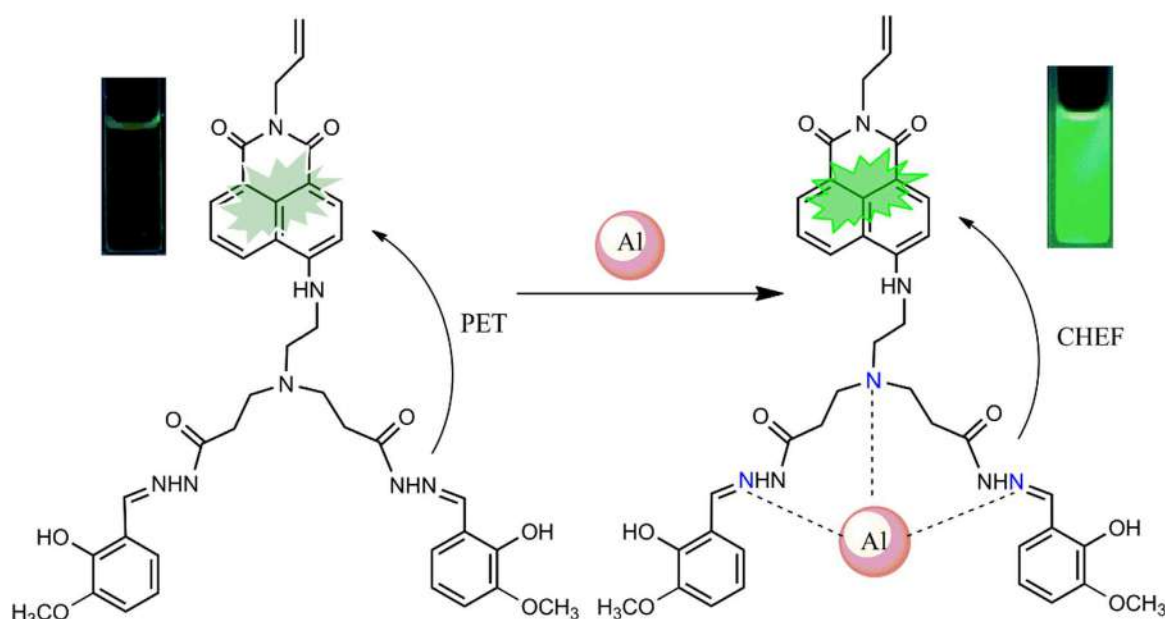


Fig. 8. Proposed mechanism of probe **NBPB** toward Al^{3+} .

Table 4

Recovery study of spiked determination of Al^{3+} in water samples.

Sample	Al^{3+} spiked (μM)	Al^{3+} recovered, mean $\pm$ SD (μM)	Recovery (%)
Yellow river water 1	4	4.11 ± 0.13	102.8
Yellow river water 2	6	6.08 ± 0.22	101.3
Yellow river water 3	8	8.04 ± 0.17	100.5
Tap water 1	4	3.95 ± 0.07	98.8
Tap water 2	6	6.03 ± 0.26	100.5
Tap water 3	8	8.19 ± 0.29	102.4

interference under neutral aqueous medium and the fluorescence intensity was insignificantly affected by counter anions of the salts of various metal ions (Fig. S7).

According to the obtained data, the binding mode is very likely due to the chelation between Al^{3+} with nitrogen atoms of the Schiff base and ethylenediamine group of **NBPB**. The proposed binding mode of the [**NBPB**– Al^{3+}] complex is illustrated in Fig. 8. As given in Fig. 8, the free **NBPB** displayed weak fluorescence, but upon the addition of Al^{3+} , [**NBPB**– Al^{3+}] exhibited a significant fluorescence enhancement at an emission maximum 528 nm.

3.5. Reversibility study

The reversibility of the recognition process of **NBPB** was performed by adding a bonding agent, Na_2EDTA . The addition of Na_2EDTA to a mixture of **NBPB** and Al^{3+} resulted in diminution of the fluorescence intensity at 528 nm, indicating the regeneration of the free sensor **NBPB** and a reversible coordination between **NBPB** and Al^{3+} (Fig. S8). Upon the addition of Al^{3+} , the fluorescence intensity of **NBPB** showed significant fluorescence enhancement at 528 nm again. The reversible response towards Al^{3+} meant that the receptor **NBPB** could be used as a selective fluorescent sensor for detection of Al^{3+} in such fields of environmental analysis and **NBPB** is a chemosensor not a chemodosimeter of Al^{3+} .

3.6. Preliminary analytical application

The practicability of probe **NBPB** was evaluated by the determination of recovery of spiked Al^{3+} in river water and tap

water samples. The water samples used were obtained from the Yellow River (Lanzhou, China) and tap water samples from the Lanzhou Jiaotong University. The environmental water samples were filtrated and showed that no Al^{3+} was present in them and then centrifuged for 15 min at 8000 r min^{-1} [53,54]. The solution of Al^{3+} at different concentrations levels of 4.00, 6.00, 8.00 μM were spiked in all real samples. The experimental results were given in Table 4. The concentration of Al^{3+} detected was close to that of the added Al^{3+} . The recovery was between 95% and 103% and further confirmed probe **NBPB** is potentially applicable for detecting Al^{3+} in practical water samples in the presence of the other coexisting ions.

4. Conclusions

In summary, we have synthesized and characterized a new probe (**NBPB**) based on naphthalimide-Schiff base that is selective for Al^{3+} . The properties of **NBPB** and the previously studied Al^{3+} probes are compared and listed in the table (Table S1). **NBPB** showed a significant fluorescence enhancement based on PET mechanism upon the addition of Al^{3+} in a solution HEPES (pH = 7.4) buffer/DMF of **NBPB**. Job's plot, fluorescence titration, UV–vis absorption titration, MS as well as NMR analysis results indicated that the interaction of **NBPB** with Al^{3+} involved the chelation method and the formation of a 1:1 complex of [**NBPB**– Al^{3+}]. The probe **NBPB** demonstrated high selectivity, sensitivity, strong interference immunity and low detection limit for detecting Al^{3+} in real environmental water samples. This work will provide inspiration for the design of new probe for the monitoring of metal ions.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.jphotochem.2016.09.004>.

CCDC 1494030 contains the supplementary crystallographic data for **NBPN**. These data can be obtained free of charge from The Cambridge Crystallographic Data Centre via www.ccdc.cam.ac.uk/data_request/cif.

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